

percentage peaks at about 50% during the period and then tapers off in 2011–2012 when markets have become increasingly competitive and some high frequency traders have exited the business. Furthermore, as we add the hedge fund trader percentage to HFT and MM, we start approaching values of about 60–70%. Thus, the classification of the different types of traders is what accounts for the disparity in statistical figures.

An important issue to point out is that the actual market volume percentage of HFT + MM + HF has only increased slightly over 2003–2012 from 57 to 70%. So, it is more likely that the more nimble market makers and hedge fund traders turned to high frequency trading strategies rather than this class of investors just appearing overnight.

Algorithmic trading is currently one of the hottest areas of capital expenditure for Wall Street firms (both buy-side and sell-side). There are numerous conferences and seminars dedicated to algorithmic trading throughout the US, Europe, Asia, and Australia. Unfortunately, the amount of academic research has not kept pace with the surge in algorithmic trading. Most industry awareness regarding algorithmic trading has come from broker-dealers whose marketing information is mainly self-serving, with the main purpose being to increase order flow and business. There is a strong need for unbiased academic research and a well-tested decision making methodology. We seek to bridge the gap between academia and Wall Street.

INVESTMENT CYCLE

The investment cycle consists of four distinct phases: asset allocation, portfolio construction, implementation, and portfolio attribution. *Asset allocation* consists primarily of distributing investment dollars across stocks, bonds, cash, and other investment vehicles in order to achieve a target level of return within a specified level of risk exposure and tolerance. *Portfolio construction* consists primarily of selecting the actual instruments to hold in each asset class. *Implementation* has historically consisted of selecting an appropriate broker-dealer, type of execution (e.g., agency transaction or principal bid), and now includes specification algorithms and algorithmic trading rules. Portfolio managers evaluate the performance of the fund to distinguish between market movement and skilled decision making ability in the *portfolio attribution* phase of the cycle.

Until more recently, the vast majority of research (academic and practitioner) has focused on improved investment decisions. Investors have a